INCH-POUND
MIL-DTL-24643/52E
22 August 2002
SUPERSEDING
MIL-C-24643/52D
22 November 1994

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, 300 VOLTS, TYPE LSTPNW

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of MIL-DTL-24643 listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation.

REQUIREMENTS:

Qualification required.

Construction (non-watertight)

First -	Copper conductor, uncoated, AWG size No. 22, class K stranding, ASTM B 172.
Second -	Separator may be used at manufacturer's option where required to provide free-stripping insulation.
Third -	Cross-linked polyethylene insulation or ethylene propylene rubber insulation 0.009 inch minimum average thickness. Telephone identification code shall be applied by method 3.
Fourth -	Two conductors cabled together to form a pair. Maximum lay 2 inches.
Fifth -	The required number of pairs (see table I), cabled with a lay not greater than 10 inches. Cabling sequence shall be consecutive, starting with pair no. 1, from the center outward. Fillers shall be used as necessary to form a firm, well-rounded assembly.
Sixth -	Binder tape applied helically with overlap.
Seventh -	Cross-linked polyolefin jacket. (see table I for thickness).

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TABLE I. Details.

Military part no. M24643/52	Size	Number of pairs	Cable Jacket Thickness min avg (inch)	Overall diameter (inches) TYPE LSTPNW min	Overall diameter (inches) TYPE LSTPNW max	Cold bending mandrel diameter (max inch)
-01UN	1-1/2	1-1/2[1/]	0.040	0.217	0.235	3.0
-02UN	3	3	0.040	0.287	0.310	3.5
-03UN	5	5	0.040	0.338	0.365	4.0
-04UN	10	10	0.040	0.435	0.470	5.5
-05UN	15	15	0.040	0.490	0.530	6.5
-06UN	20	20	0.040	0.532	0.575	7.0
-07UN	30	30	0.050	0.629	0.680	8.0
-08UN	40	40	0.050	0.708	0.765	9.0

1/ Size -1/2 is composed of three conductors cabled to form a triad. Colored coded black, white, and red.

EXAMINATION AND TESTS:

Basic electrical:	Requirements
Conductor Resistance - ohms/1000 feet at 25°C, maximum	17.94
Voltage withstand - volts, root mean square, minimum	
Conductor to conductor	2000
Insulation resistance - Conductor to conductor,	
Megohms -1000 feet, minimum	1000
Conductor continuity	No failure

Group A:	
Visual and dimensional	No failure

Group B:	
Cross-linked proof test (percent, maximum)	
Insulation	50
Jacket (When tested at 200°C)	50
Cold bending, cable - (see table I for mandrel diameter)	No damage
Tear - pounds per inch thickness, minimum (ASTM D 470)	35
Physicals (unaged)	
Insulation	
Ethylene propylene rubber	
Tensile strength - lb/in^2, minimum	700
Elongation - percent, minimum	250

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Cross-linked polyethylene
Tensile strength - lb/in^2, minimum 1800
Elongation - percent, minimum 250
Jacket (cable)
Tensile strength - lb/in^2, minimum 1300
Elongation - percent, minimum 160

Group C:
Physicals - (aged)
Insulation
Ethylene propylene
Air oven
Tensile strength - percent of unaged, minimum 75
Elongation - percent of unaged minimum 75
Cross-linked polyethylene
Air oven
Tensile strength - percent of unaged, minimum 80
Elongation - percent of unaged minimum 80
Jacket (cable)
Air oven
Tensile strength - percent of unaged, minimum 60
Elongation - percent of unaged minimum 60
Hot oil immersion
Tensile strength - percent of unaged, minimum 50
Elongation - percent of unaged, minimum 50
Shrinkage ..... No failure
Heat distortion - percent of unaged, maximum 30
Permanence of printing (jacket) - cycles, minimum 125

Group D:
Flame propagation (cable) ..... No failure

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B, C and D, plus the following:

	Requirements
Aging and compatibility (cable) (125 ± 5°C)	No failure
Abrasion resistance (jacket) - scrapes, minimum	75
Acid gas equivalent - percent, maximum	
Jacket	2
Fillers	2
Insulation	18

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Halogen content - percent, maximum
- Jacket 0.2
- Fillers 0.2

Immersion (jacket)
- Tensile strength - percent of unaged, minimum 50
- Elongation - percent of unaged, minimum 50

Smoke index, maximum
- Jacket 25
- Fillers 45
- Insulation 45

Toxicity index, maximum
- Jacket 5
- Fillers 5
- Insulation 1.5

Durometer (jacket) - (type A) hardness, minimum 80
Weathering (jacket ..... No failure

UNIT ORDERING LENGTHS:

Type and size	Feet (nominal)
LSTPNW-1-1/2 through -10	2500
LSTPNW-15 and -20	1500
LSTPNW-30 and -40	1000

Custodians:
Army - MI
Navy - SH

Preparing Activity:
Navy - SH
(Project 6145-2308-050)

Review Activities:
Army - AV, CR
Navy - CG, EC
DLA – CC

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